

## BLOCKCHAIN EXPERIMENT - 8

**AIM:** To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

### THEORY:

**1. Blockchain Technology** Blockchain is a distributed ledger technology that maintains a continuously growing list of records (blocks), each cryptographically linked to the previous one. Since every node in the network holds an identical copy of the ledger, no single party can alter records without consensus from others, making it inherently tamper-proof. Core characteristics include decentralization (no central authority), immutability (records cannot be modified post-confirmation), transparency (all participants can view transactions), and security via cryptographic hashing algorithms like SHA-256.

**2. Ethereum and Smart Contracts** Ethereum extends basic blockchain functionality by enabling programmable logic through smart contracts — autonomous pieces of code deployed on-chain that execute automatically when specific conditions are triggered. These contracts are authored in Solidity, a statically-typed language, then compiled to EVM (Ethereum Virtual Machine) bytecode that runs identically across all network nodes. Every execution consumes *gas*, a fee mechanism that compensates validators and prevents resource abuse. Each deployed contract receives a unique address, and the `msg.sender` global variable identifies the calling account at runtime.

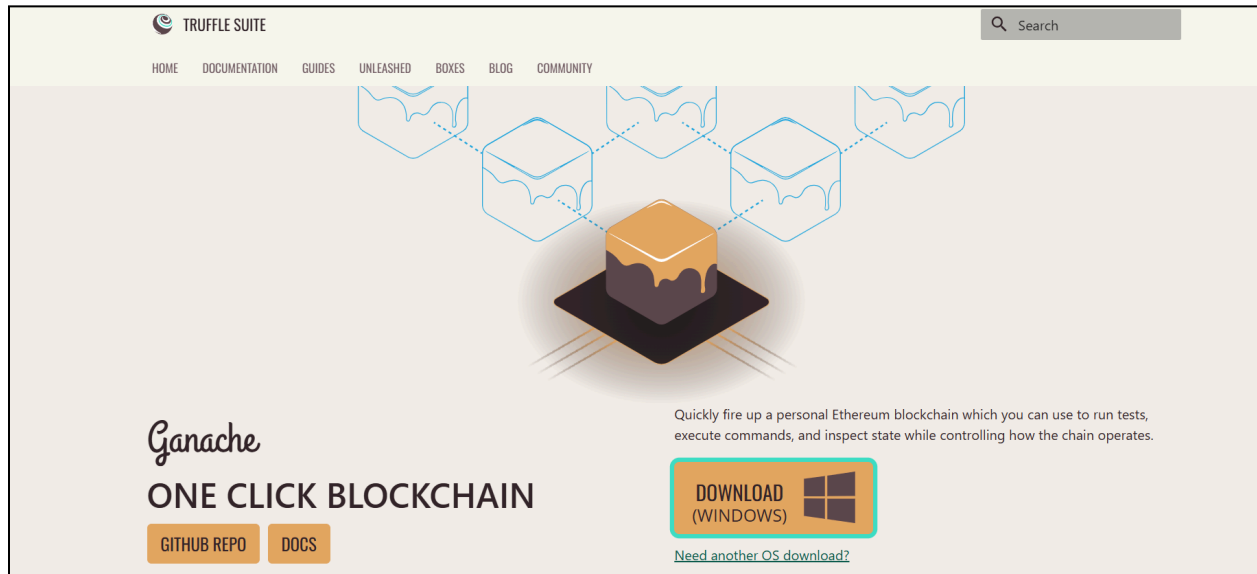
**3. Ganache** Ganache is a local Ethereum blockchain emulator primarily used during development and testing phases. It eliminates the need to interact with public networks by spinning up a private chain on the developer's machine, complete with pre-funded test accounts (10 accounts × 100 ETH each). Transactions are mined instantly, and its GUI provides real-time visibility into account balances, block data, and transaction logs. It exposes a local RPC endpoint (typically <http://127.0.0.1:7545>) that development tools can connect to seamlessly.

**4. MetaMask** MetaMask is a browser-based Ethereum wallet that bridges web applications and the blockchain. It manages account credentials, signs outgoing transactions, and routes them to whichever network it is configured for — in a development context, this is pointed to the local Ganache instance rather than the Ethereum mainnet, enabling cost-free testing with simulated ETH.

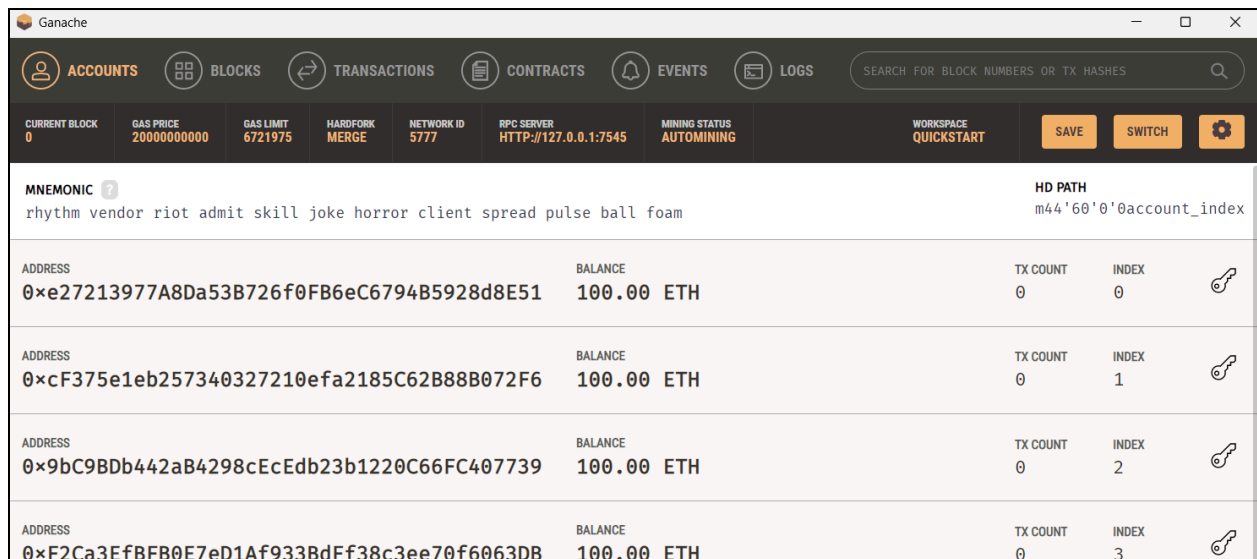
**5. Remix IDE** Remix is a browser-accessible development environment tailored for Solidity smart contract development. It bundles a code editor, compiler, deployment interface, and debugger into a single tool. By selecting the *Injected Provider* environment option, Remix delegates transaction signing to MetaMask, forwarding operations to whichever network MetaMask is currently connected to.

## IMPLEMENTATION:

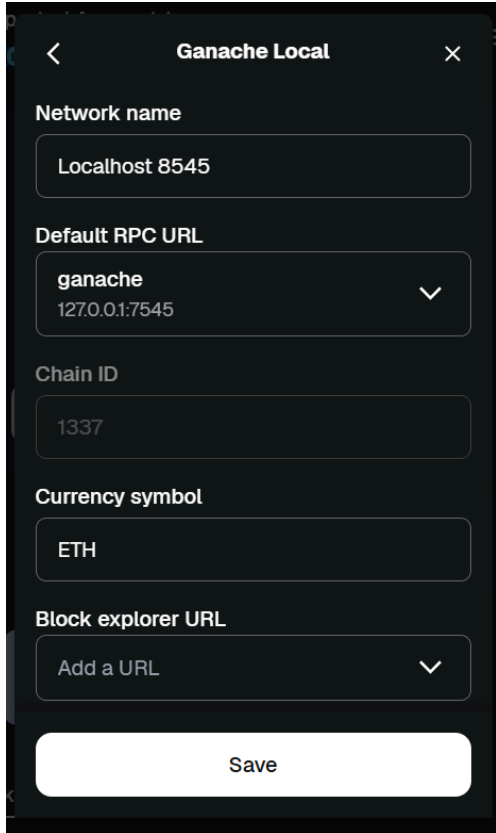
### Step 1: Download and install ganache



Once installed, open ganache and click on Quickstart Ethereum to start localhost.

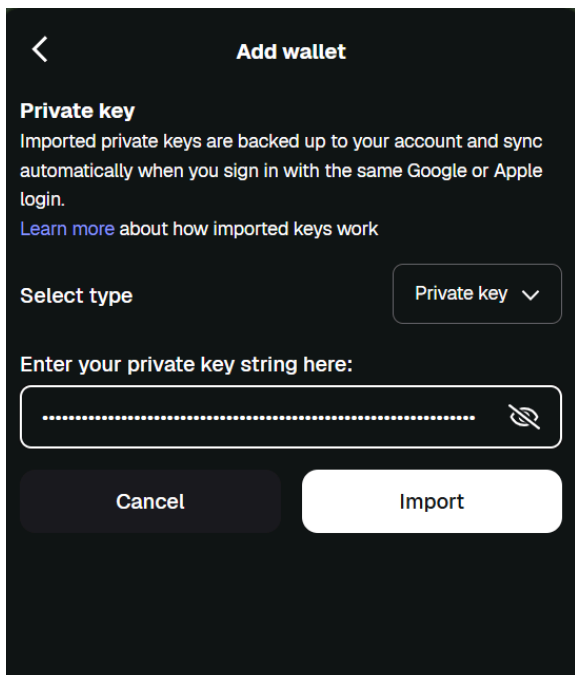


**Step 2: In MetaMask, click on the networks tab and add a custom network.**

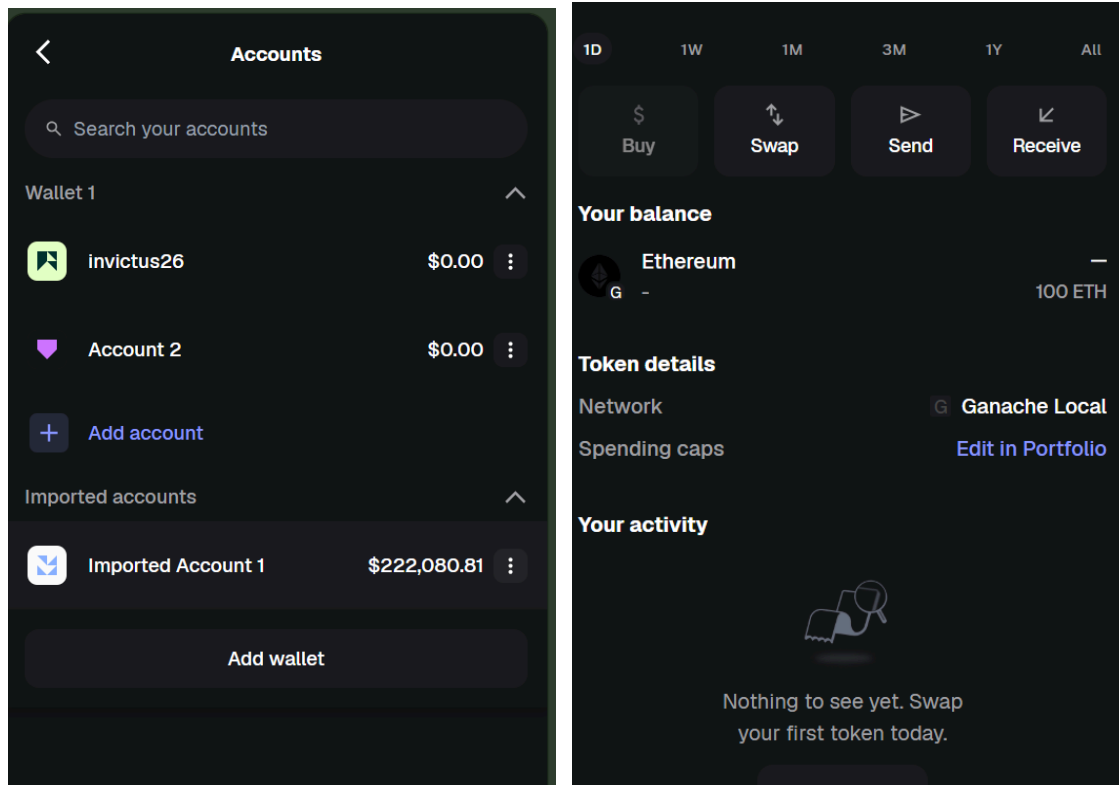


The screenshot shows the 'Add network' screen in MetaMask. At the top, there is a title bar with a back arrow, the text 'Ganache Local', and a close button. Below this, there are several input fields: 'Network name' with the value 'Localhost 8545', 'Default RPC URL' with a dropdown menu showing 'ganache' and '127.0.0.1:7545', 'Chain ID' with the value '1337', 'Currency symbol' with the value 'ETH', and 'Block explorer URL' with a dropdown menu showing 'Add a URL'. At the bottom, there is a large white button labeled 'Save'.

**Step 3: Import ganache account into metamask**



The screenshot shows the 'Add wallet' screen in MetaMask. At the top, there is a title bar with a back arrow and the text 'Add wallet'. Below this, there is a section titled 'Private key' with a description: 'Imported private keys are backed up to your account and sync automatically when you sign in with the same Google or Apple login.' and a link 'Learn more about how imported keys work'. Below this, there is a 'Select type' dropdown menu with the value 'Private key'. Below the dropdown, there is a text input field labeled 'Enter your private key string here:' with a placeholder text '.....' and a toggle icon. At the bottom, there are two buttons: 'Cancel' and 'Import'.

**Wallet balance appears in metaMask account**

**Step 4:** In Remix IDE, create a new file **crowdfunding.sol**

**Code:**

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;

contract Crowdfunding {
    address public owner;
    uint public goal;
    uint public deadline;
    uint public totalFunds;

    mapping(address => uint) public contributions;

    constructor(uint _goal, uint _duration) {
        owner = msg.sender;
        goal = _goal;
        deadline = block.timestamp + _duration;
    }
}
```

```
}

function contribute() public payable {
    require(block.timestamp < deadline, "Deadline passed");
    require(msg.value > 0, "Send some ETH");

    contributions[msg.sender] += msg.value;
    totalFunds += msg.value;
}

function withdrawFunds() public {
    require(msg.sender == owner, "Only owner");
    require(totalFunds >= goal, "Goal not reached");

    (bool success, ) = payable(owner).call{value: totalFunds}("");
    require(success, "Transfer failed");
}

function refund() public {
    require(block.timestamp > deadline, "Not ended");
    require(totalFunds < goal, "Goal was reached");

    uint amount = contributions[msg.sender];
    require(amount > 0, "No contribution");

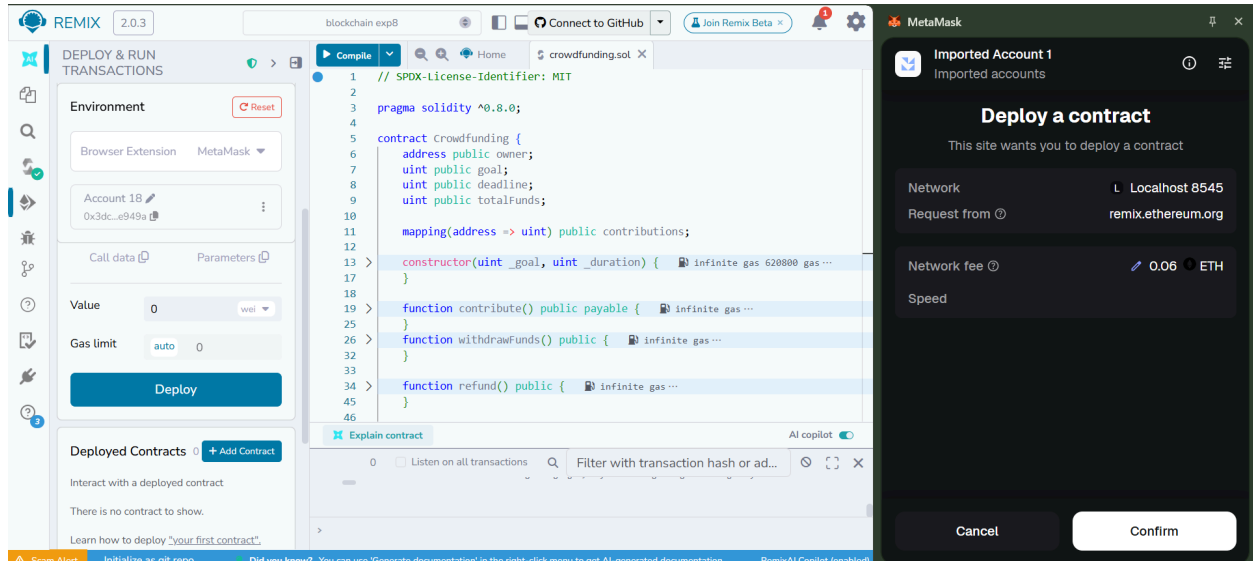
    contributions[msg.sender] = 0;

    (bool success, ) = payable(msg.sender).call{value: amount}("");
    require(success, "Refund failed");
}

function getBalance() public view returns (uint) {
    return address(this).balance;
}
}
```

**Step 5: Compile and run the contract.**

When prompted in metamask, confirm the transaction.



The contract is deployed!

The deducted amount can be seen in ganache app along with transaction count

The screenshot shows the Ganache application interface. At the top, there are tabs for ACCOUNTS, BLOCKS, TRANSACTIONS, CONTRACTS, EVENTS, and LOGS. Below the tabs, a table lists account details. The table has columns for ADDRESS, BALANCE, TX COUNT, and INDEX. The first account (index 0) has a balance of 99.92 ETH and a TX COUNT of 2. The subsequent accounts (indices 1-5) have a balance of 100.00 ETH and a TX COUNT of 0.

| ADDRESS                                    | BALANCE    | TX COUNT | INDEX |
|--------------------------------------------|------------|----------|-------|
| 0x678aB28b947030404778C6193423B73c7a014b18 | 99.92 ETH  | 2        | 0     |
| 0xcC163421b1091647e96D5Ce9ddEdF1e624dad0a6 | 100.00 ETH | 0        | 1     |
| 0x7Ea6E7307F61C1b29eA2FC7c40F16e602b514a31 | 100.00 ETH | 0        | 2     |
| 0xb00C1B0A9e7E90b2b538A98d7DD98A499578F4dd | 100.00 ETH | 0        | 3     |
| 0x176239f7D57783b2Fe396cb7FA3B73178C750713 | 100.00 ETH | 0        | 4     |
| 0x9BFCFd4b9223c4d5d551bb8152DBAFFAF449A36B | 100.00 ETH | 0        | 5     |

**CONCLUSION:**

Ganache provides a simple and powerful environment to simulate a local Ethereum blockchain. By integrating it with **Metamask** and **Remix**, developers can deploy, test, and debug smart contracts efficiently before moving to a public network.